MANIPAL UNIVERSITY COLLEGE MALAYSIA

25 YEARS

Histology Lecture

Thyroid and parathyroid glands

Dr. Madhan



for instruction purpose only ..

#inspire

Lecture outline

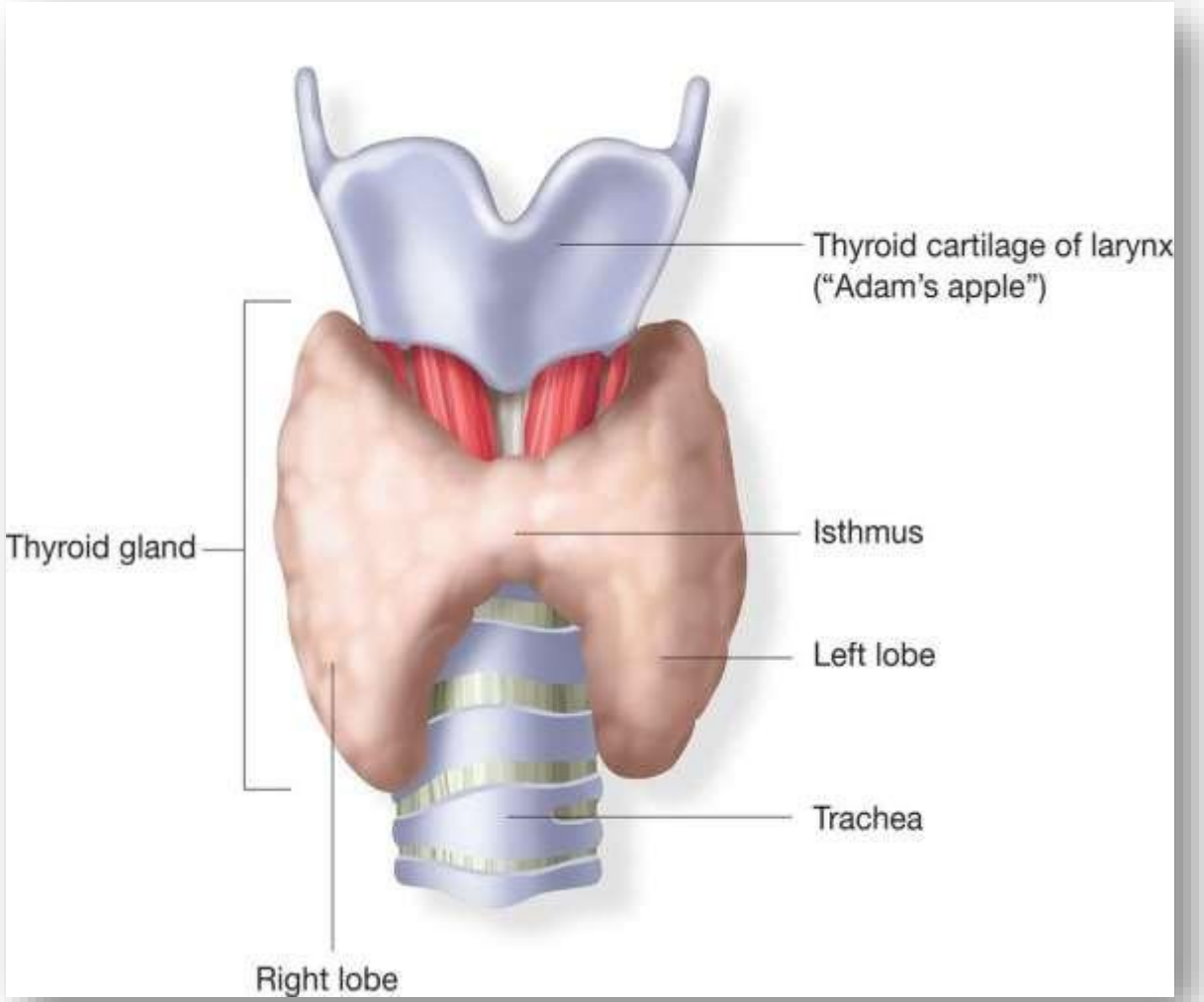
At the end of the lecture, students should be able to:

- Thyroid
  - Illustrate the microscopic structure of thyroid gland

- Parathyroid
  - Illustrate the microscopic structure of parathyroid gland

Thyroid gland

- Endocrine gland which is consisting of 2 lobes that are connected by isthmus.
- It lies against vertebrae C5-C7 and T1; occupying the upper part of trachea.
- Highly vascular.
- **Functions:** Produce the thyroid hormone and calcitonin hormone.

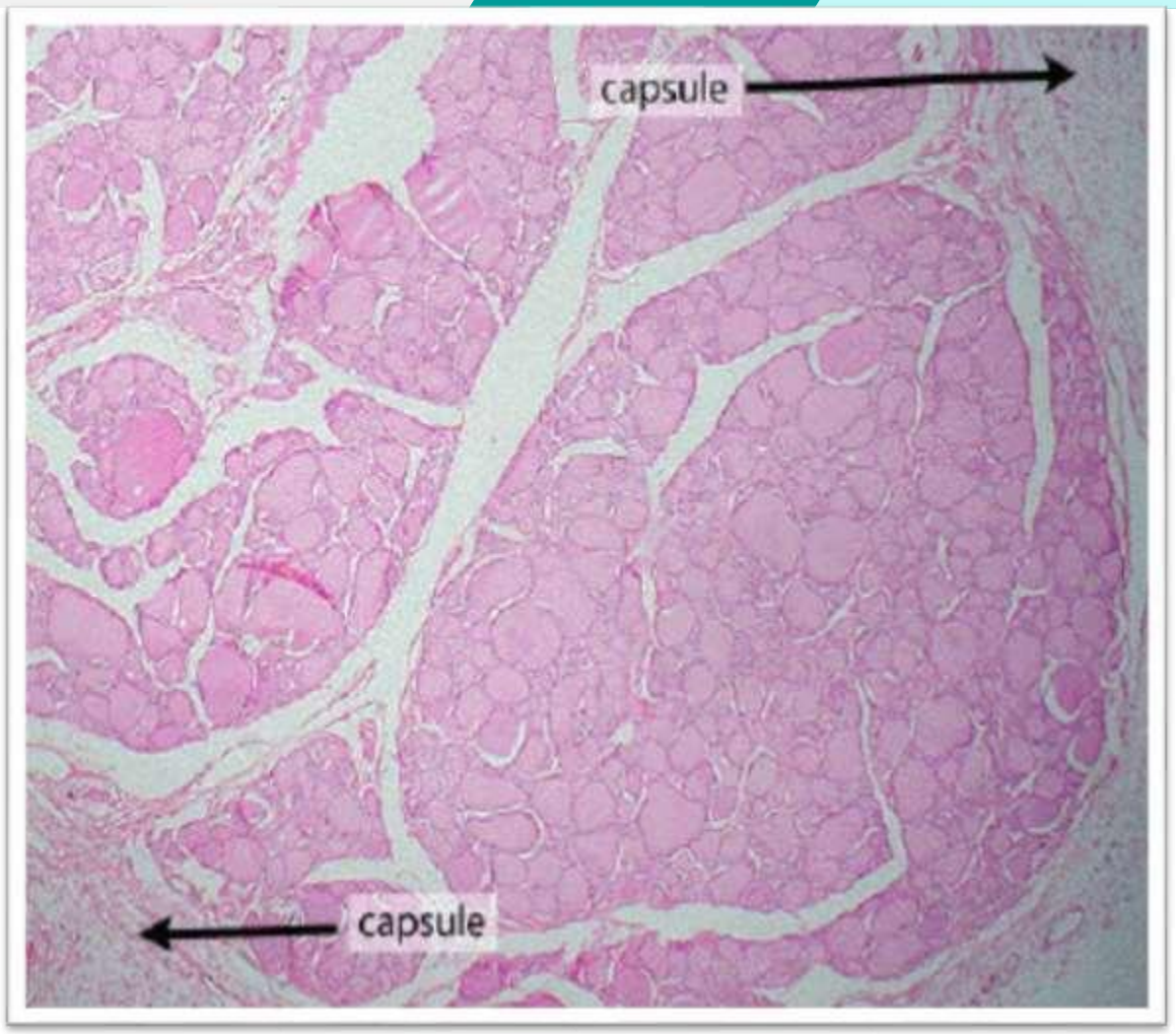


Capsule covering

- Thyroid gland is surrounded by connective tissue capsule.

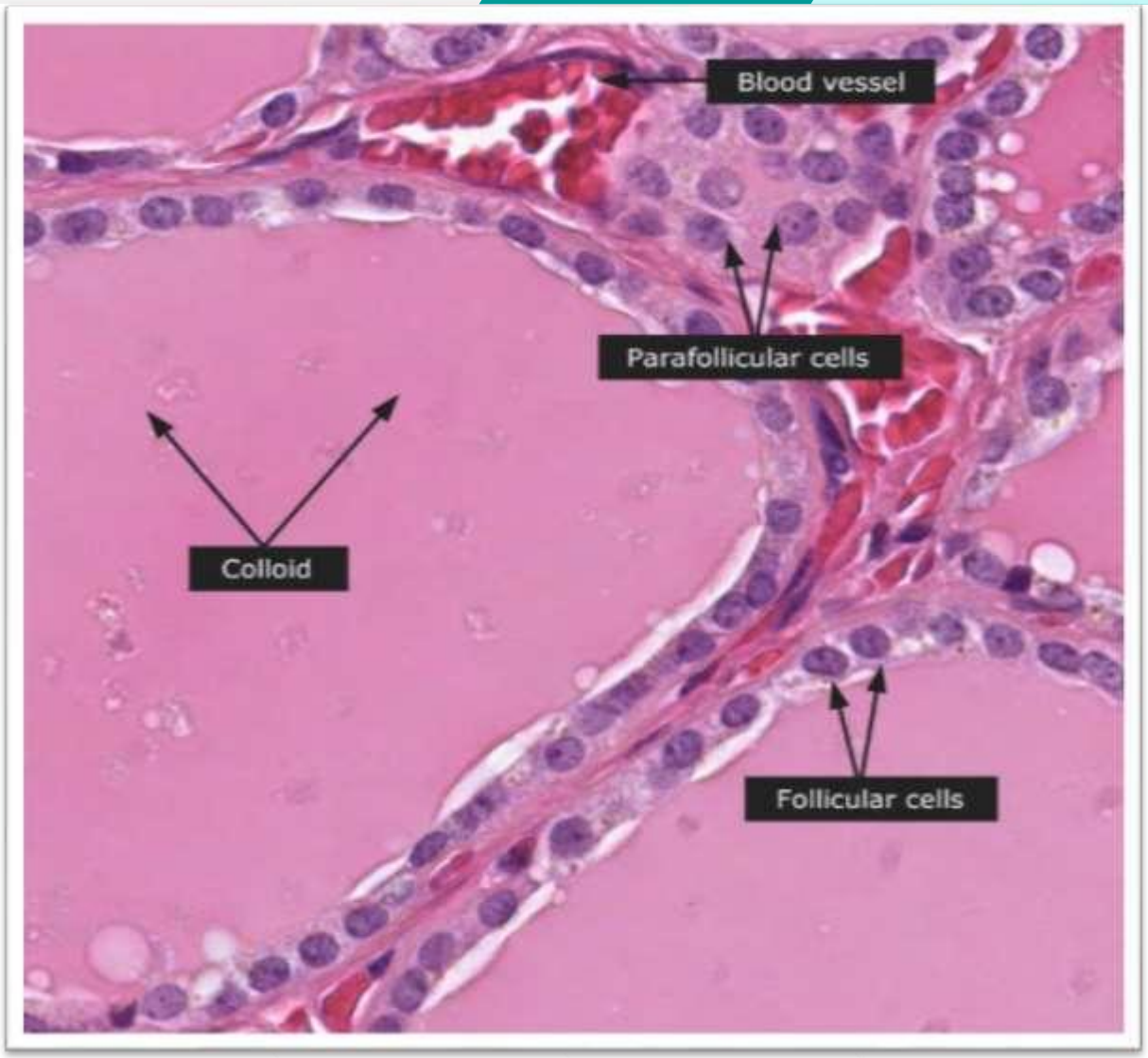
- Septa extend from the capsule into the glands and divide into lobules.

- Septa is also important as passage for blood vessels, lymphatic vessels and nerve fibers.



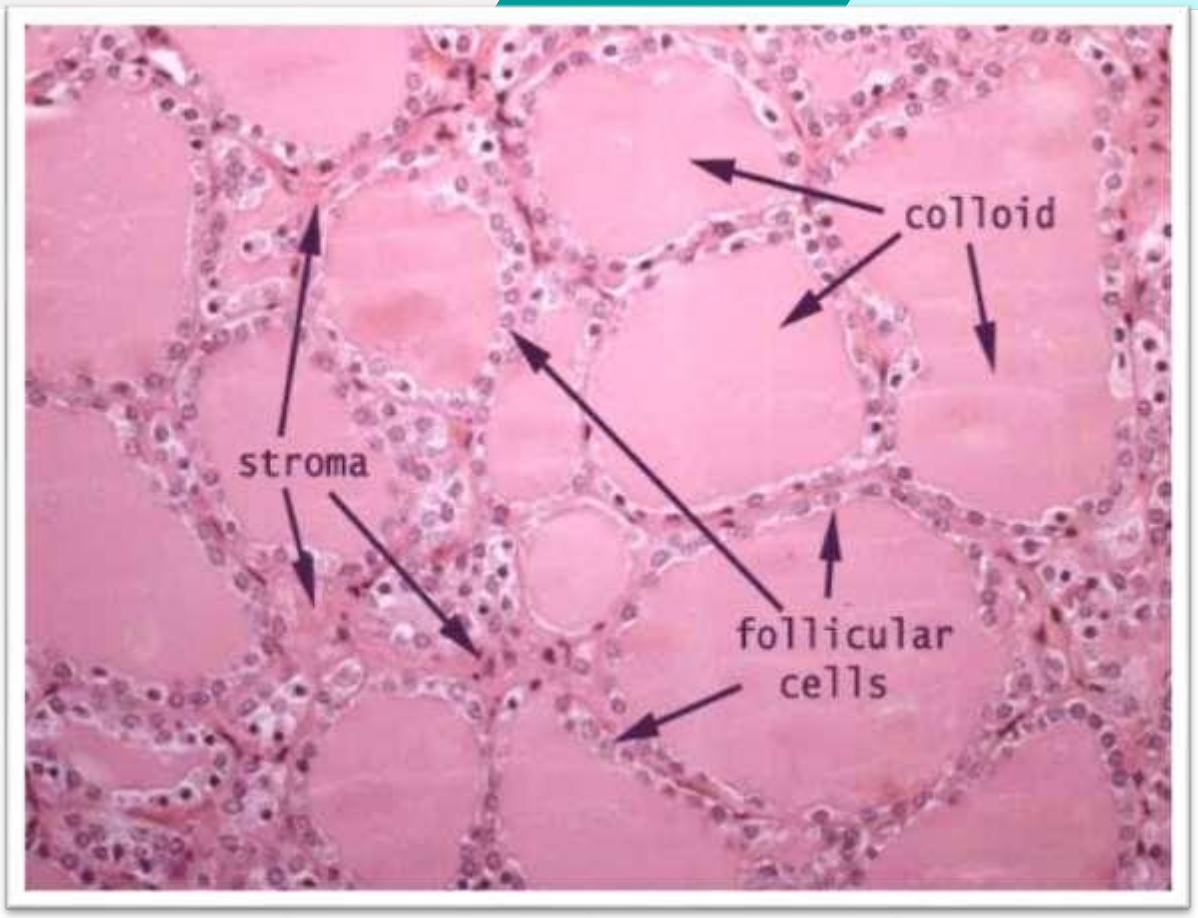
Histological structures

- Thyroid follicle is the structural & functional unit of the thyroid gland.
- Each lobule contain several thyroid follicle.
- Each follicle consist of a single layer of epithelial cells called **follicular cells** and a central lumen filled with a gelatinous substance called **colloid.**
- **Parafollicular cells (C cells)** are also found scattered in the thyroid follicle.

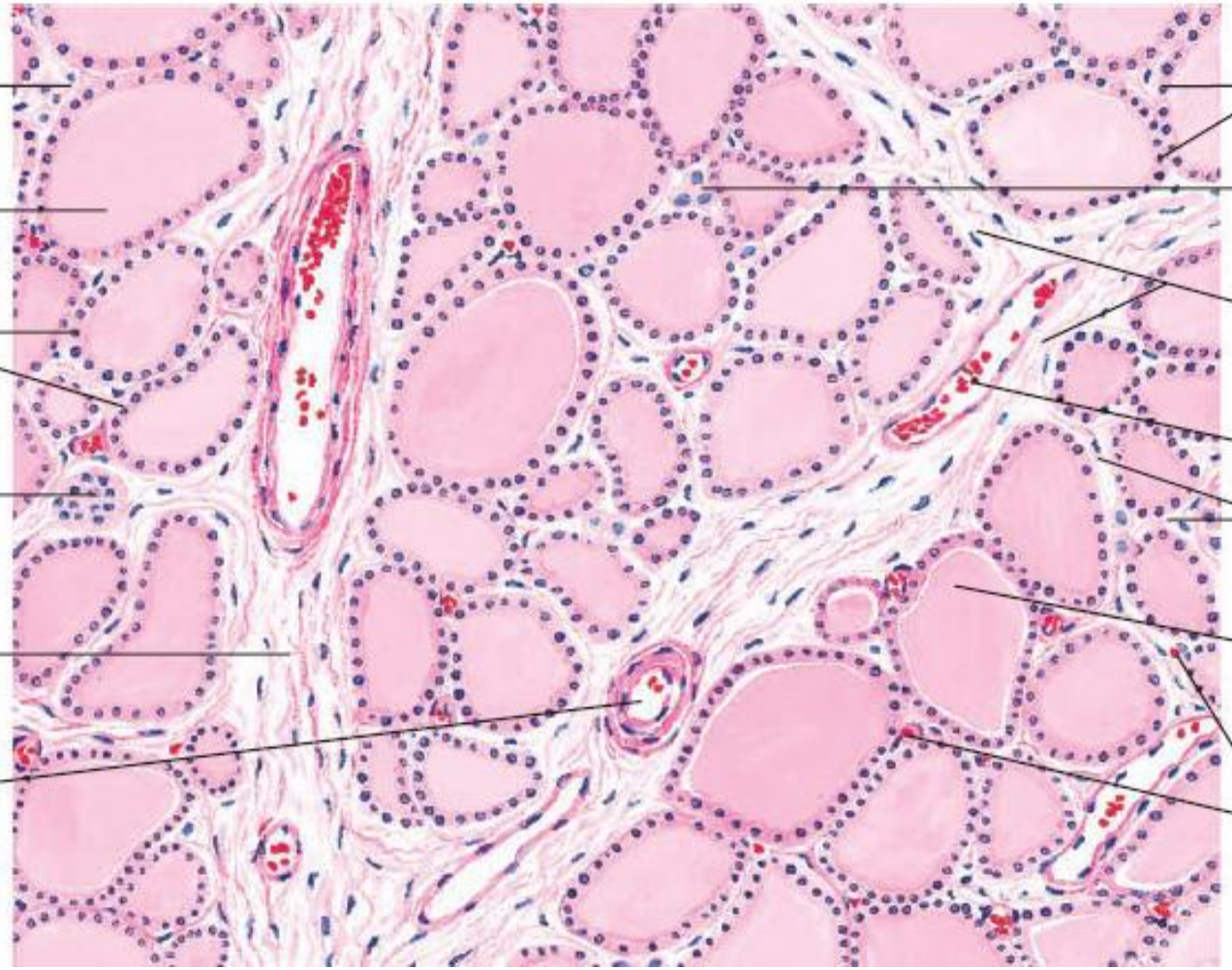


Follicular cells

- Ranging from simple squamous (inactive) to cuboidal (moderately active) to columnar epithelium when stimulated (Highly active).
- Cells are rich in rER and Golgi complexes; nucleus is round and centrally located in the cell.
- Contain many small apical vesicles which is involved in transport & release of thyroglobulin protein into the colloid.
- Possess short, blunt microvilli that extend into the colloid.



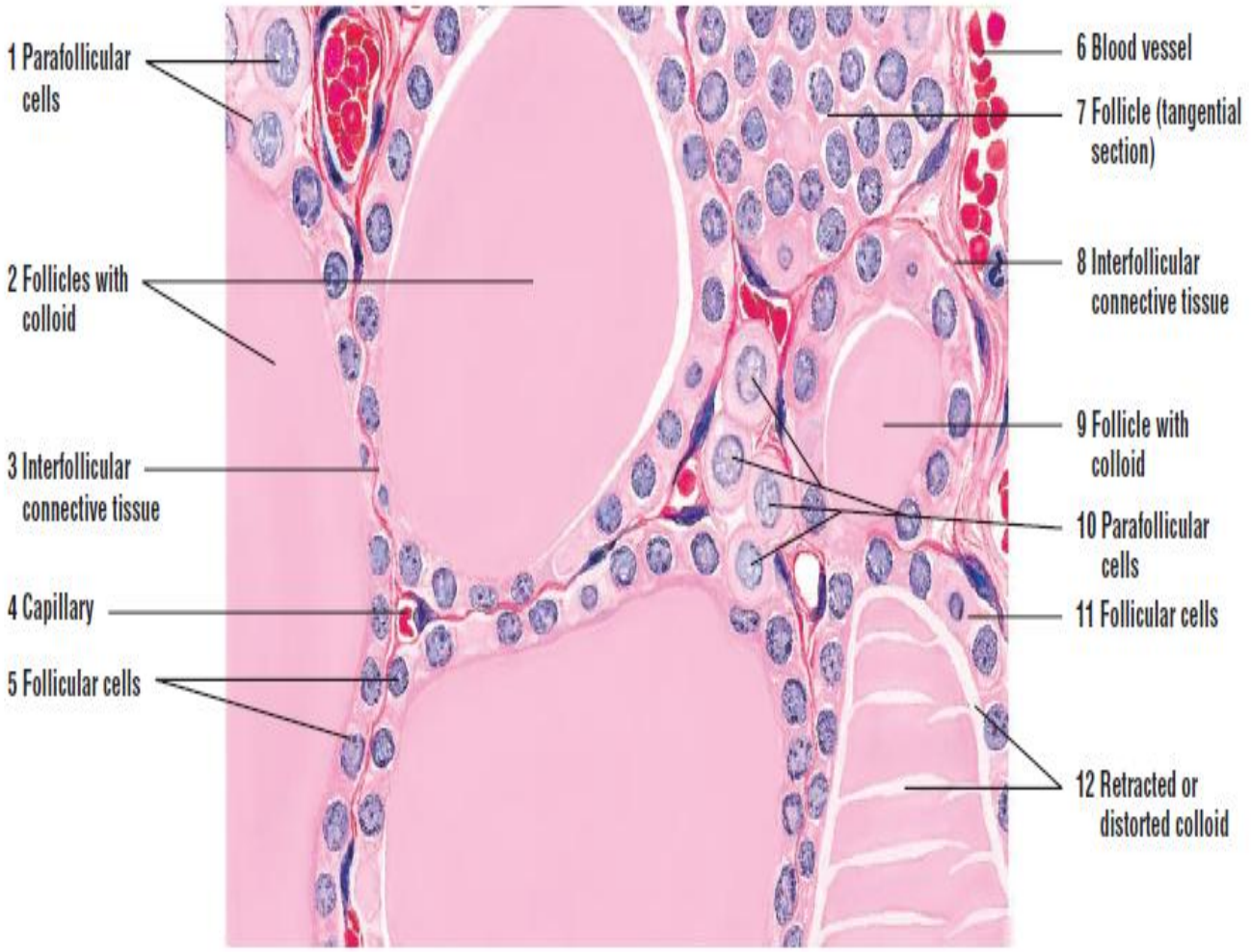
Function: Synthesis and release of **thyroid hormones; Thyroxine (T_4)** and **triiodothyronine (T_3)**


1 Parafollicular cells
2 Follicle with colloid
3 Follicular cells
4 Follicle (tangential section)
5 Connective tissue septa
6 Arteriole
7 Follicular cells
8 Parafollicular cells
9 Connective tissue septa
10 Venule
11 Interfollicular connective tissue
12 Follicle with retracted colloid
13 Capillaries

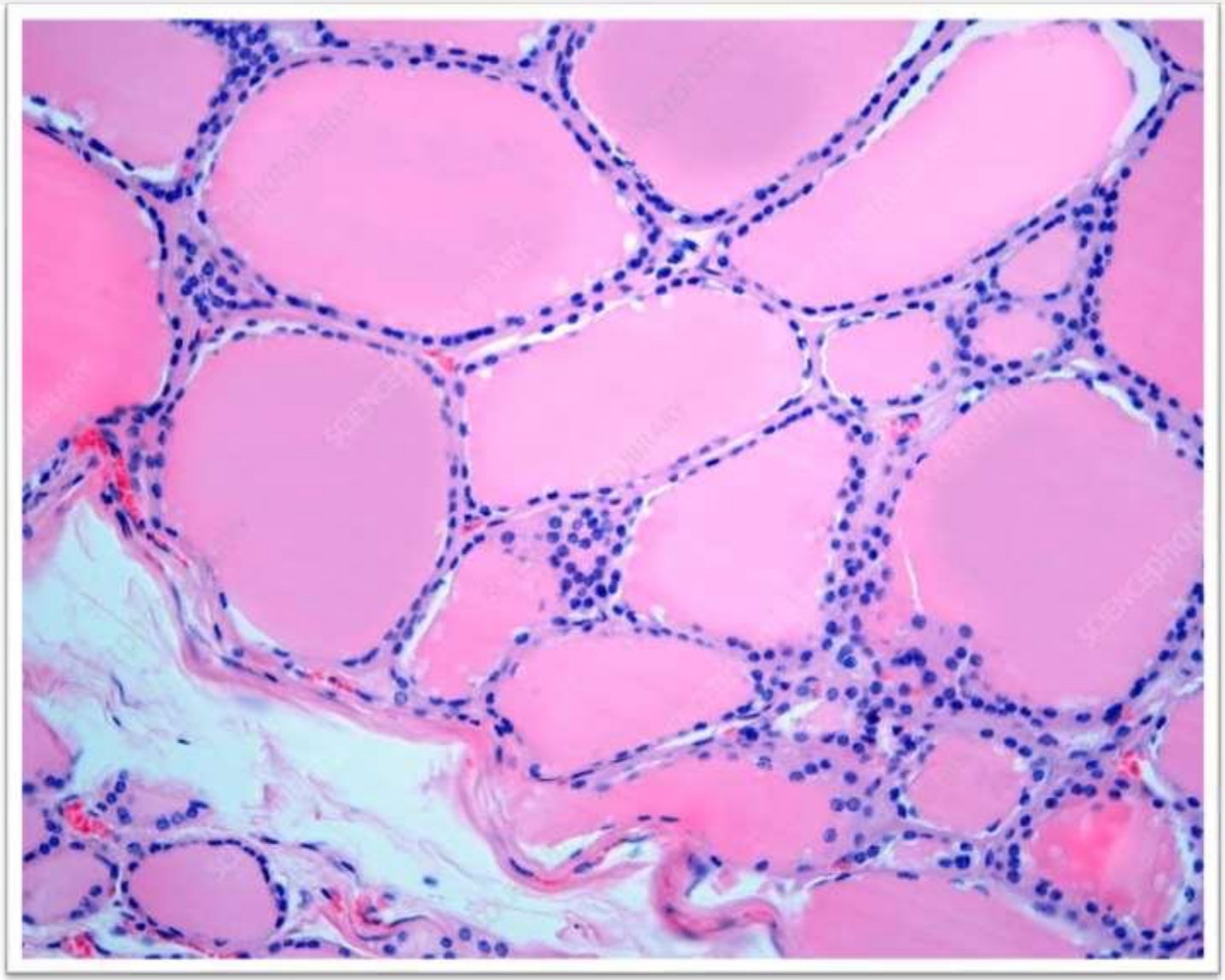
Parafollicular cells (C cells)

- Polyhedral shape cell with oval eccentric nuclei; Larger and stain less intensely than follicular cells.
- Present singly or in small clusters of cells located between follicular cells and basal lamina; may also be found scattered between follicles.
- Do not reach the lumen.
- Have small amount of rER, large Golgi complexes and numerous small granules containing polypeptide hormone.

Function: Synthesis and release of **calcitonin**

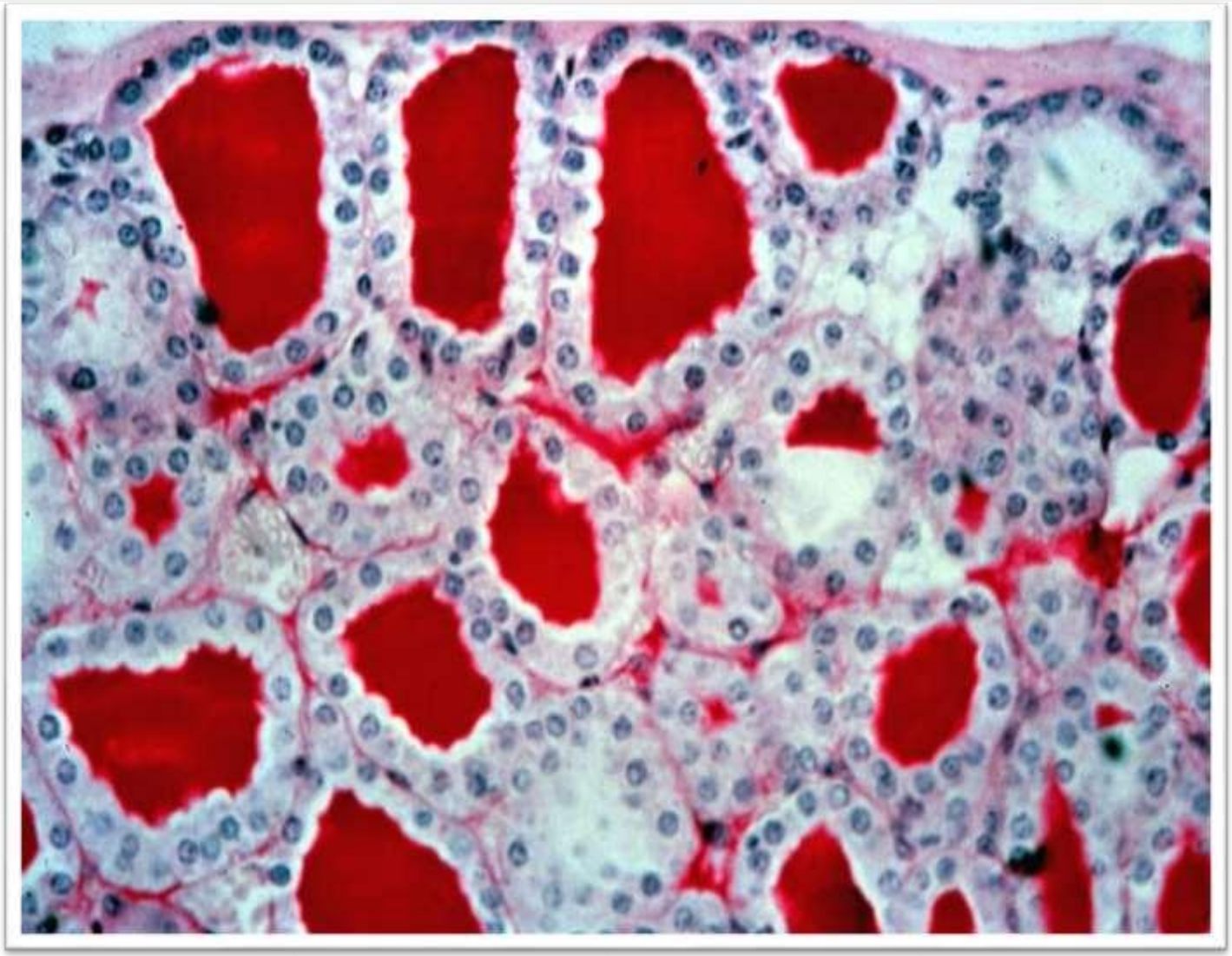


Hypoactive thyroid follicles



- The follicles shown here appear hypoactive.
- Follicular epithelium is flattened.
- Colloid is abundant.

Hyperactive thyroid follicles



- The follicles shown here appear to be hyperactive.
- The follicular cells appears particularly large.
- Colloid substance in the follicles is depleted, suggesting it is being taken up rapidly for conversion to thyroid hormones

Medical application

Hyperthyroidism

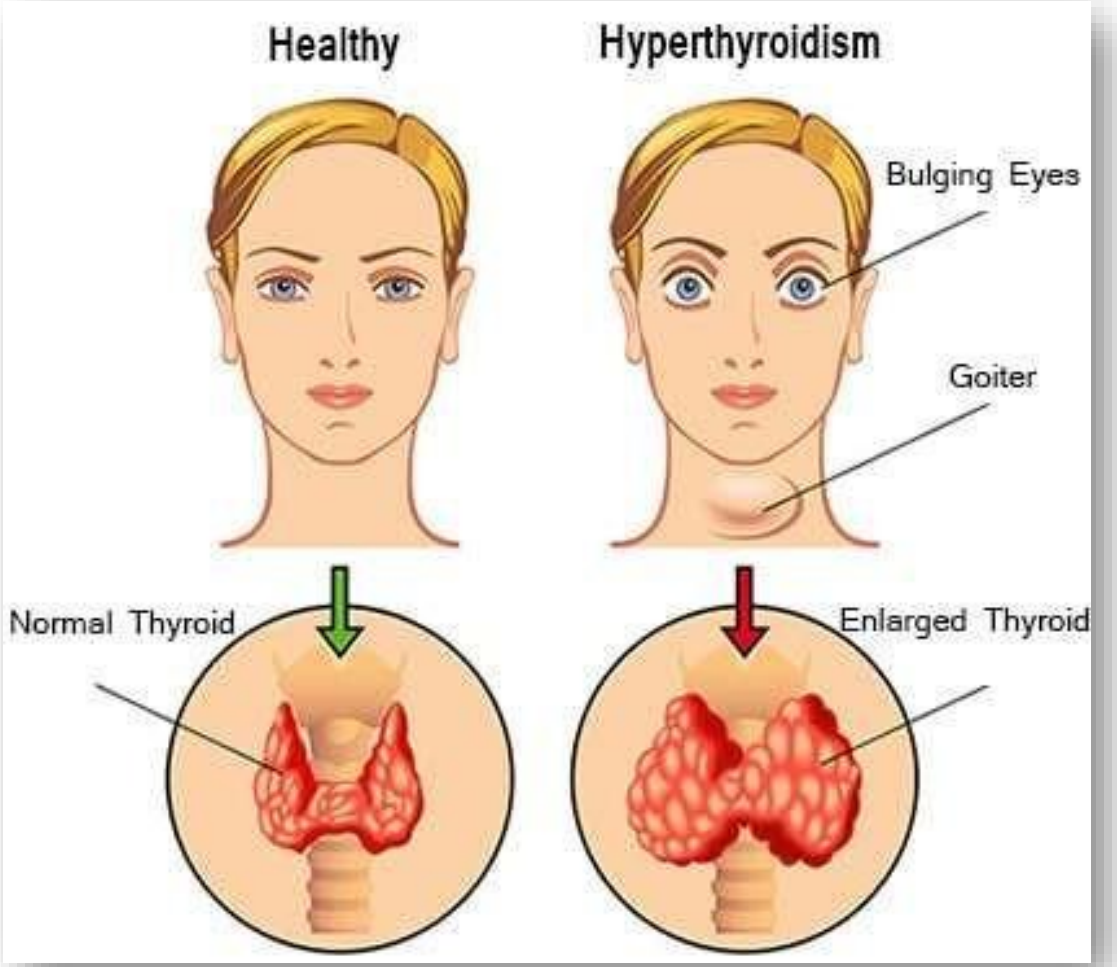
- Overactivity of the thyroid gland resulting in excess production of thyroid hormone by the gland.

- Thyrotoxicosis is the clinical manifestations resulting from the hyperthyroidism condition.

- The condition is common in middle aged women.

- Diffusely enlarged gland.

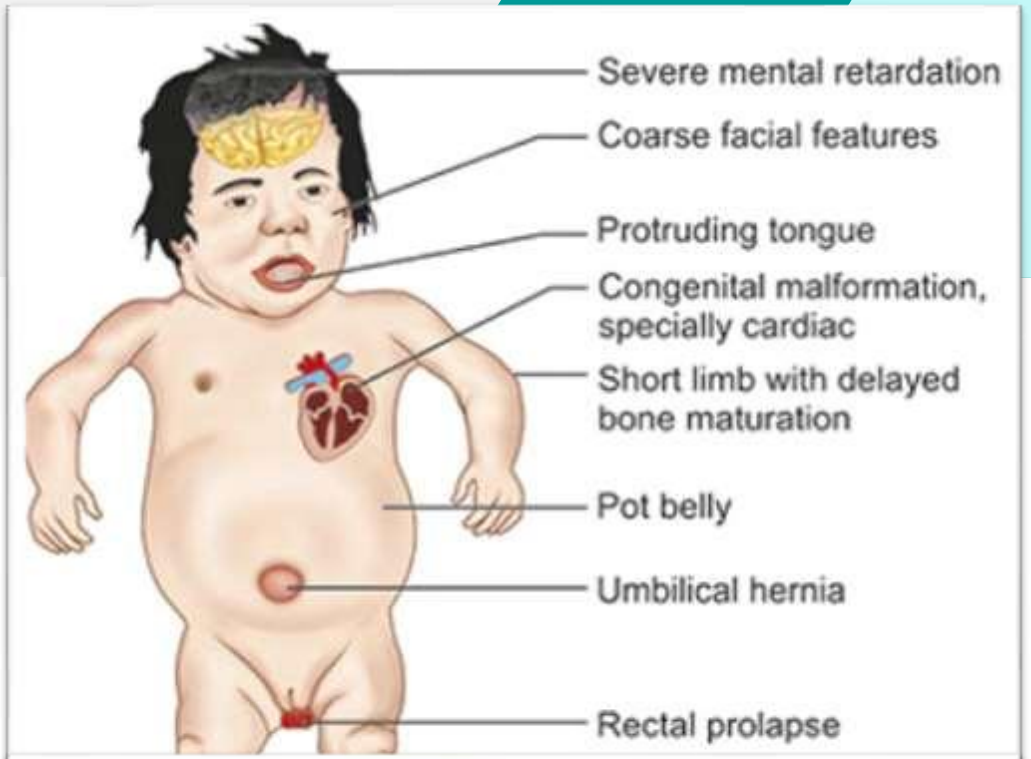
- The walls of thyroid follicles are much in folded and the epithelial cells lining the follicular walls are taller than normal.

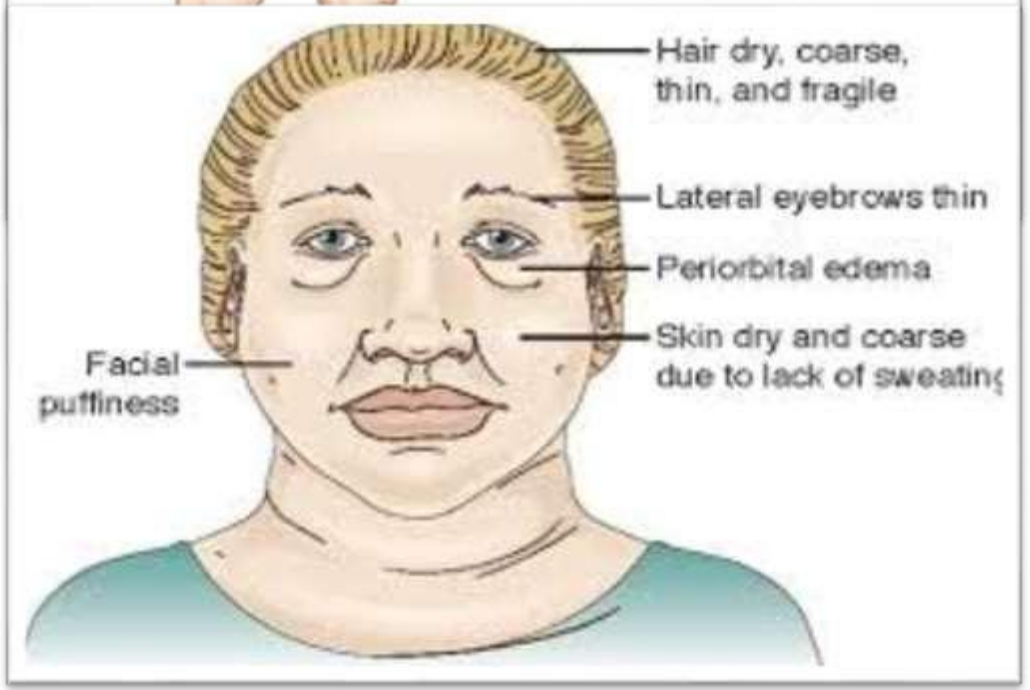


Hypothyroidism

1) Congenital hypothyroidism (Cretinism) in infants:
   - Due to failure of development of thyroid gland.
   - The child is born with normal appearance but showing the signs of retarded mental and physical growth.

2) Myxedema in adults:
   - Due to spontaneous degeneration of thyroid gland as resulting from hypopituitarism, hypothalamic failure, or excessive treatment of hyperthyroidism by drugs/surgery.
   - Common in female; Slowing of mental and physical processes, increase in body weight.





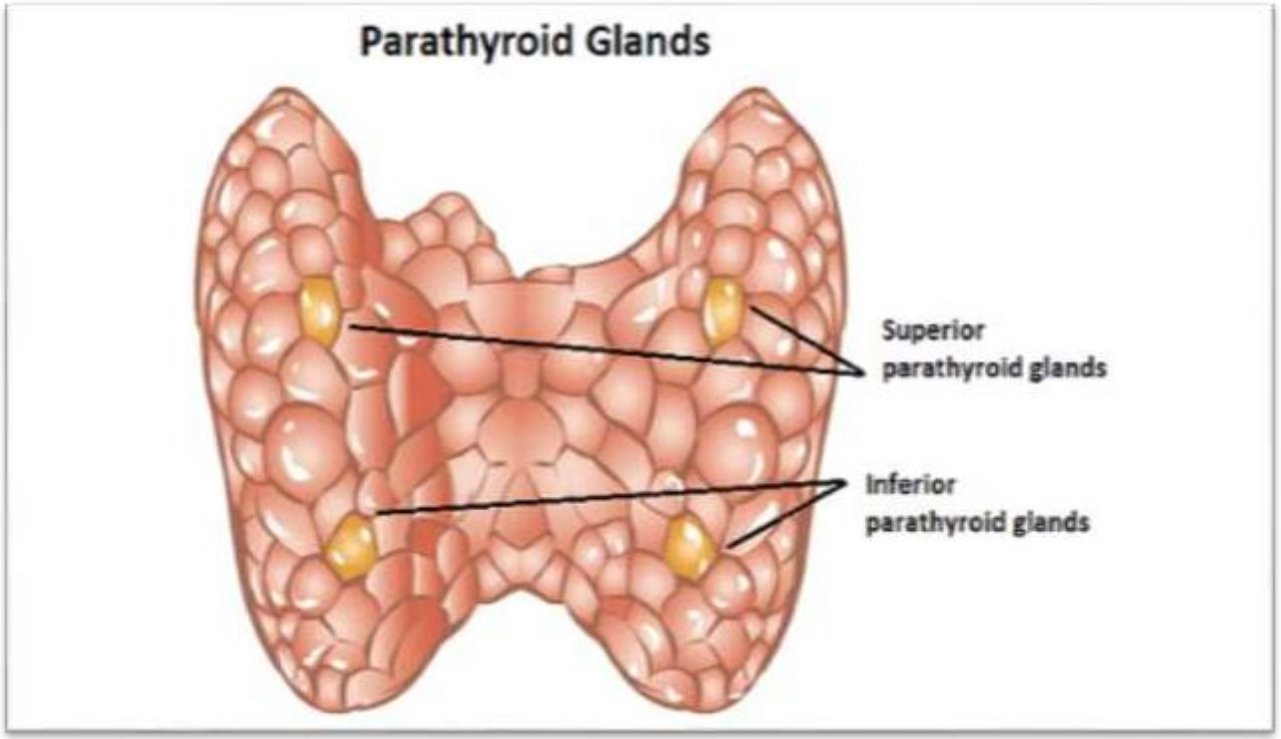
Parathyroid gland

- Four oval masses of small endocrine glands located on the posterior surface of thyroid gland, usually embedded in its capsule.

- One gland at each end of the upper and lower pole.

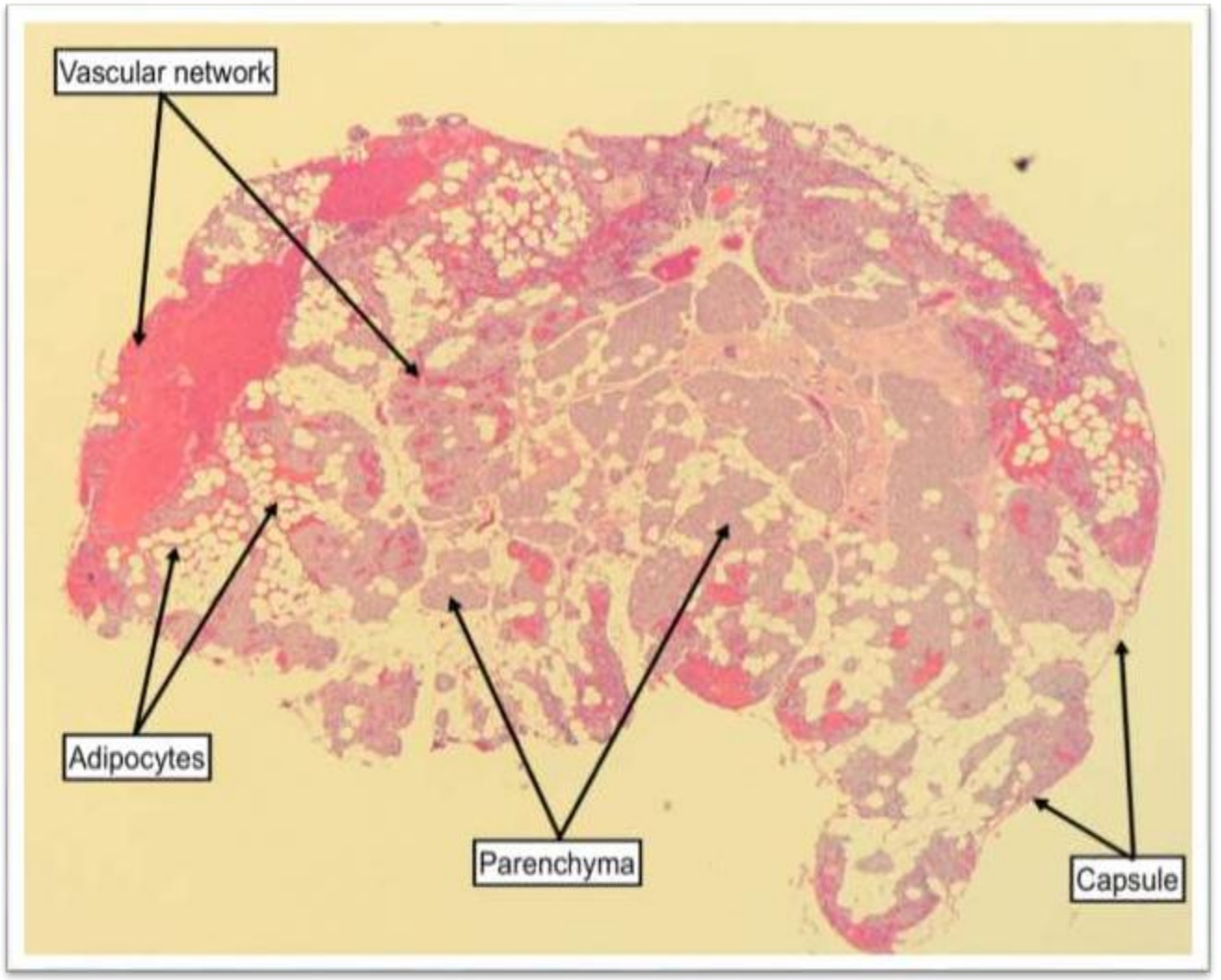
- Accessory parathyroid gland may be found in mediastinum or in the neck.

- **Functions:** Produce the parathyroid hormone (PTH)



Capsule covering

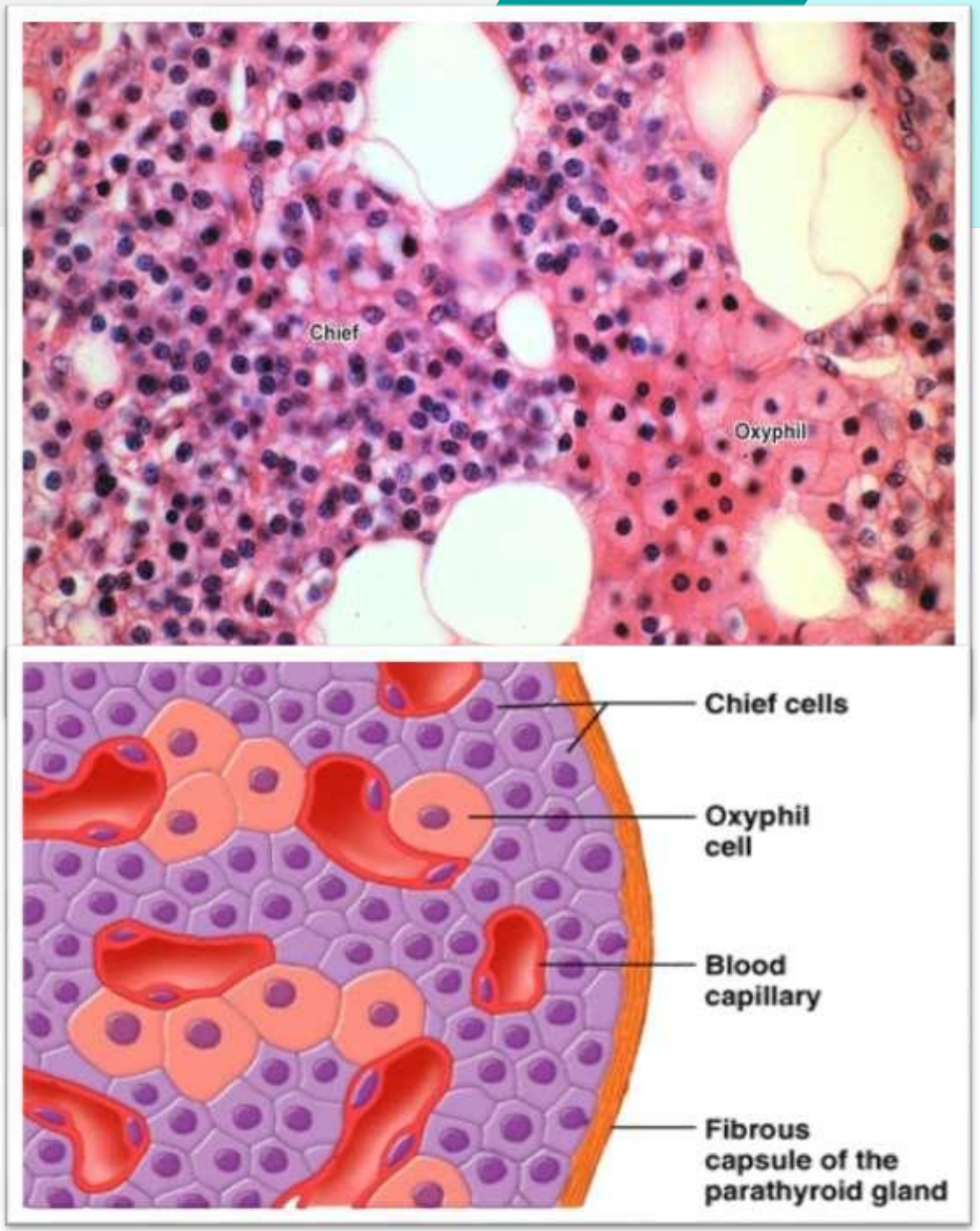
- Each gland is contained in a capsule which send septa into the gland.
- Septa will merge with the reticular fibers that support that secretory cells.
- With increasing age, many secretory cells are replaced by adipocytes.



Histological structure

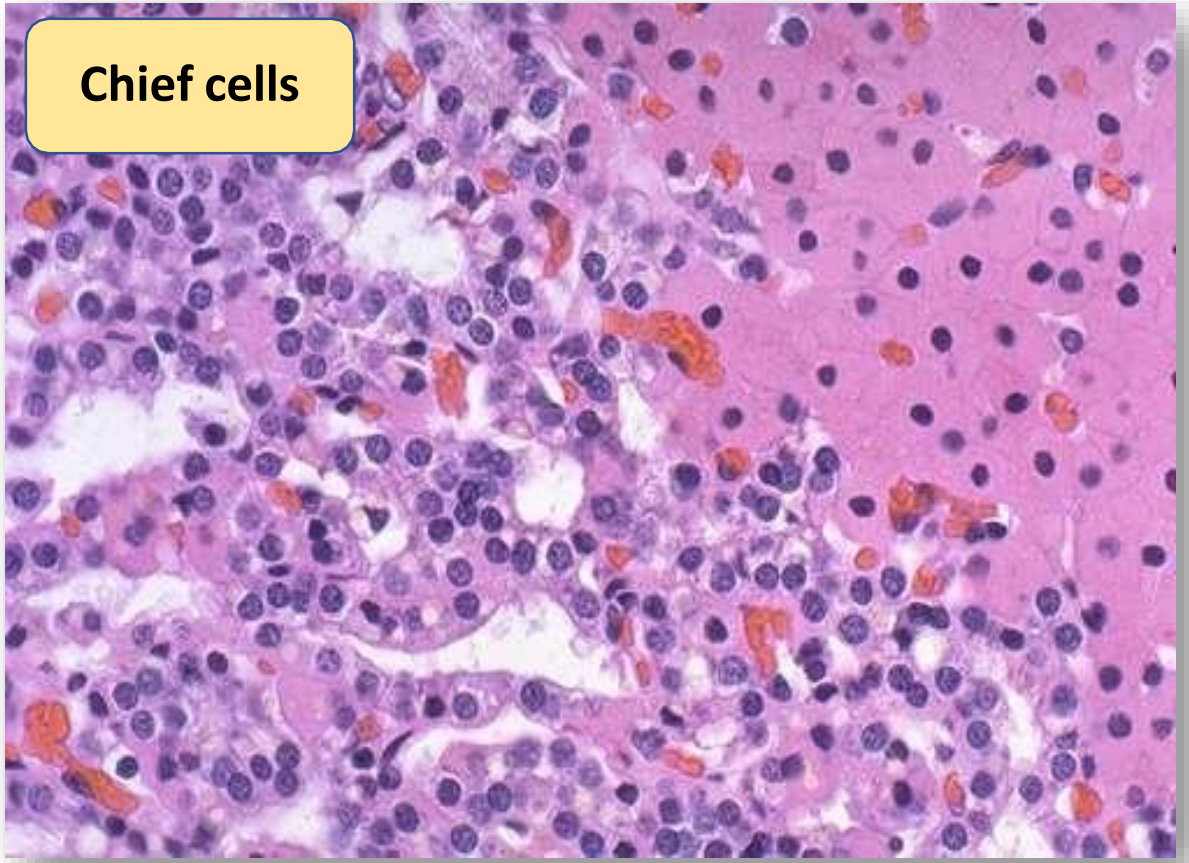
- The glands consists of two types of cells:
  - ✓ Chief (principal) cells
  - ✓ Oxyphil (eosinophil cells)

- These cells are arranged in cord-like or follicular-like clusters.



Chief (principal) cells

- Small and basophilic cells.
- Polygonal in shape arranged in clusters.
- They form anastomosing cords, surrounded by a rich fenestrated capillary network.
- Possess abundant RER, well-developed Golgi complex & secretory granules of variable size.
- **Function:** Synthesize and secrete **parathyroid hormone (PTH)** which acts on bone & kidney to raise calcium level in serum.

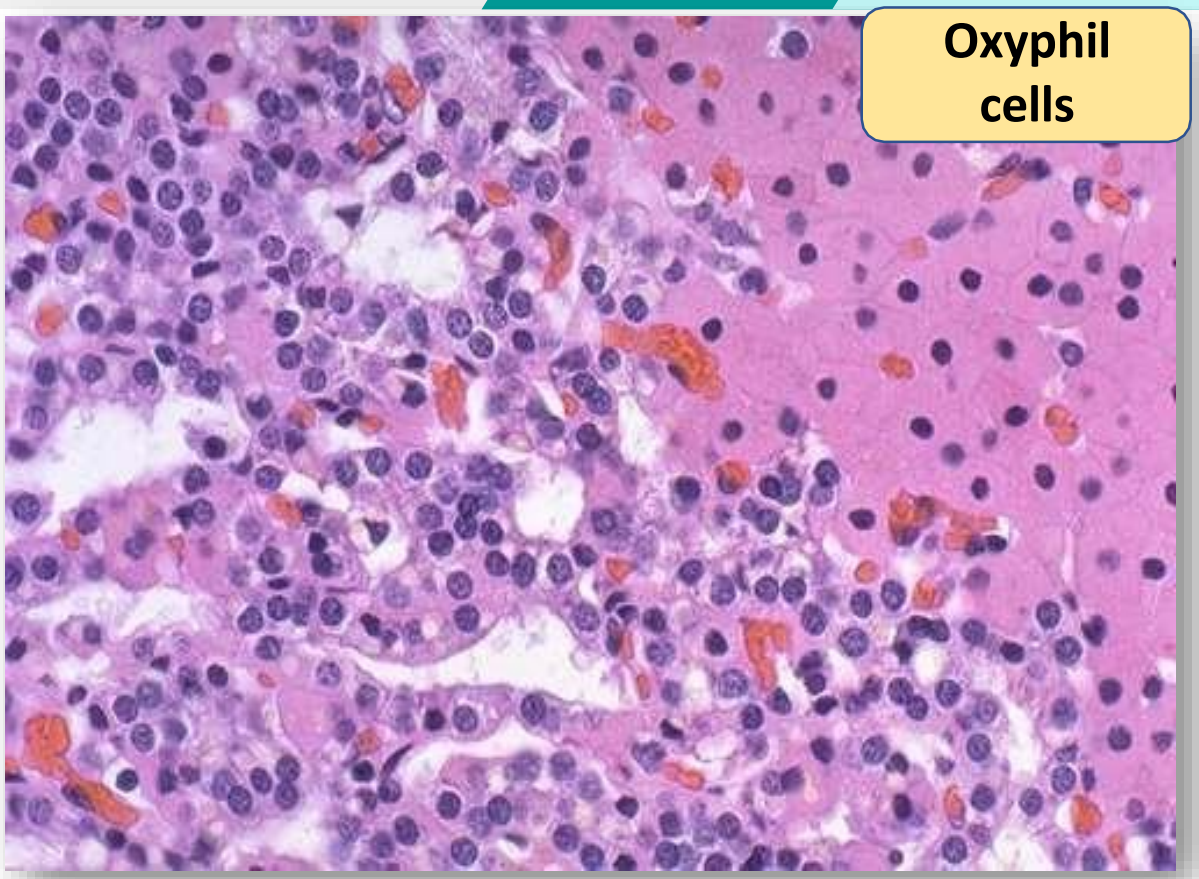


Oxyphil (eosinophil) cells

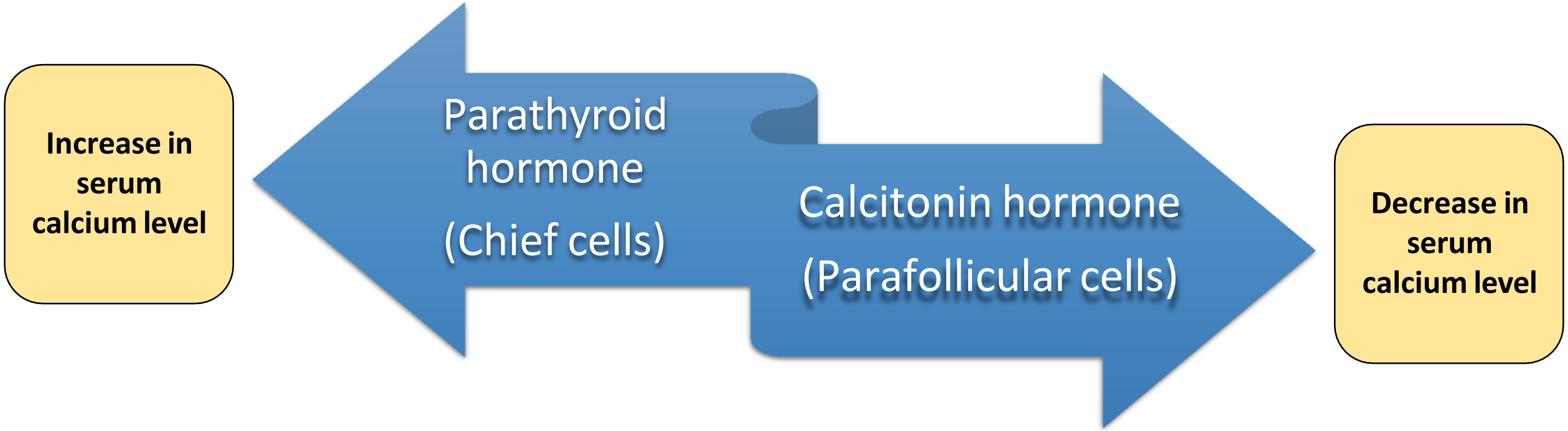
- Large, eosinophilic cells that present singly or in small clusters within the parenchyma of the gland.

- Possess abundant mitochondria, have poorly developed Golgi complex and limited amount of RER.

- Function: Unknown (may represent transitional chief cells)



Regulation of serum calcium



THANK YOU !